

Time-Series Linkage of Korea's HS codes Using Synthetic ID

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Abstract

The analysis of trade statistics suffers from time-series discontinuities arising from the frequent divergence and convergence of Harmonized System (HS) codes. To overcome this limitation, Pierce and Schott (2009) proposed the synthetic product code methodology as a groundbreaking alternative. However, to the best of the author's knowledge, no domestic or international paper or report has applied their methodology to the HSK (Harmonized System of Korea). To fill this gap, this paper constructs a consistent time-series concordance table by applying the synthetic product code methodology of Pierce and Schott (2009) to the entire universe of South Korea's 10-digit HSK product codes over the 17-year period from 2010 to 2026. The implementation code is publicly released on GitHub in both Python and Stata versions. To enable broad accessibility, the resulting concordance is further packaged as an executable (.exe) application that offers interactive visual querying and is publicly released on the same GitHub repository, providing subsequent researchers with a reference resource for micro-level time-series analyses of Korean trade data.

Introduction

In the empirical time-series analysis of international trade data, the most fundamental yet potentially fatal source of error is the temporal discontinuity of the product classification code — the Harmonized System (HS). The HS code, administered by the World Customs Organization (WCO), was introduced to classify globally traded goods under a single unified standard, but it undergoes sweeping revisions on a roughly five-year cycle in order to reflect shifts in the international trade environment, the emergence of new products driven by technological progress, and the retirement of existing items whose trade volumes have become negligible. Beyond this, national statistical and customs authorities subdivide the six-digit international standard to capture their own trade in fine detail and to apply tariff rates: the United States operates ten-digit Schedule B (exports) and HTS (imports) codes, while South Korea operates the ten-digit HSK (Harmonized System of Korea), revising it annually or once every few years.

This continual updating of the classification has the merit of accurately reflecting the economic reality of each period, but it produces serious statistical artifacts for researchers who seek to build long-run time series in order to estimate industrial evolution or the effects of trade policy. The most frequent problem is the "convergence" and "divergence" of existing product codes. The essence of this problem can be grasped through a simple example. Suppose that, as of 2020, the

HSK code for apples is A, for oranges is B, and for peaches is C, and that in 2021 the Korea Customs Service consolidates all three into the single code A (convergence) and abolishes B and C. In this case, even though the actual volume of apple imports has not changed at all, the 2021 import value recorded under HSK code A appears to rise sharply relative to 2020, because the trade values of oranges and peaches have been folded into code A. Conversely, if in 2019 apples, oranges, and peaches were all contained in the single code A and were then split into codes A, B, and C in 2020 (divergence), the trade value of code A in 2020 appears to collapse relative to the previous year, because the volumes of oranges and peaches have moved out into B and C.

Comparing the import and export values, weights, or unit prices of a particular HS code over time on the basis of such naive linkage alone is therefore a hazardous approach, because it becomes impossible to distinguish whether a statistically observed surge or plunge in trade value reflects a genuine change in market demand or the effect of a trade agreement, or is merely an administrative byproduct of the customs authority's reorganization of the classification system. Leading U.S. government research institutes and scholars have found that such time-series discontinuities lead to a serious overestimation of changes in the extensive margin—that is, the entry and exit of products. According to statistics produced by Pierce and Schott (2009) (hereafter, P&S) using their own synthetic product code concordance, over the sixteen years from 1989 to 2004, as of 2004, 49% of U.S. import products and 40% of export products had undergone at least one HS code change, and these products with a history of code changes accounted for 62% of U.S. import value and 48% of export value in that same year. This implies that, if the time-series discontinuity of the classification system is left uncontrolled, nearly half of U.S. trade value can be distorted by statistical artifacts. Building further on this, Bernard, Jensen, Redding, and Schott (2009) reestimated the U.S. extensive margin for 1993–2003 using this P&S concordance and showed that, over short-run one-year intervals excluding the U.S. recession (2000–2002), on average 76% of the change in export value originated from time-series discontinuities in HS codes and 25% from the addition or discontinuation of products and trading partners, while the net contribution of firm entry and exit itself was negligible. In short, this empirically confirmed that, without controlling for changes in the classification system, the micro-level extensive margin is overestimated, whereas once these changes are controlled for, its share is far smaller than the literature had intuitively assumed.

The methodology now most widely adopted in the literature to address this problem at its root is precisely the "synthetic product code (Synthetic ID)" technique devised by P&S. This technique traces every HS code that has passed through convergence or divergence over the analysis period, binds them into a single large "family tree," and assigns each family a unique synthetic product code. Applying this technique to the fruit example above, so long as apples, oranges, and peaches have any history of crossing over—whether through convergence or divergence—the three items are merged into a single synthetic group across the entire 2019–2021 period. As a result, the abnormal, abrupt increases or decreases in import value that arise when items are merged or split in a given year can be effectively offset.

Against this theoretical background, this paper constructs a concordance table for South Korea's ten-digit HSK codes from 2010 to 2026 using the P&S methodology. To this end, this paper surveys application cases in domestic academic journals and government-funded research institute reports to confirm the gap in the prior literature, traces the development of the HS linkage techniques that have been discussed in the international literature, and, ultimately,

provides an in-depth analysis of the inherent limitations of the P&S technique and of the recent research trends aimed at overcoming them, so as to offer a solid methodological foundation for future studies analyzing South Korea's HSK.